DEPARMENT OF INFORMATION TECHNOLOGY
FACULTY OF ENGINEERING AND TECHNOLOGY

IT24-425 P-II
COMPUTER NETWORKS
MORNING

LAB#2

TASK: DESIGN OF LOCAL AREA NETWORK

Aim: Study of network IP and Connect the computers in Local Area Network.

Following is required to be study under this practical.

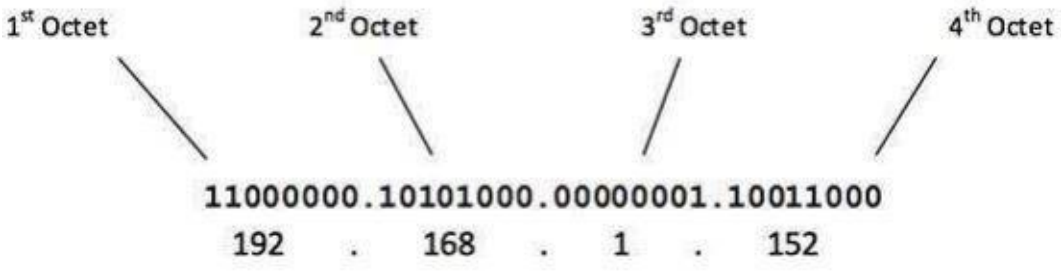
- Classification of IP address

As show in figure we teach how the ip addresses are classified and when they are used.

Class	Address Range	Supports
Class A	1.0.0.1 to 126.255.255.254	Supports 16 million hosts on each of 127 networks.
Class B	128.1.0.1 to 191.255.255.254	Supports 65,000 hosts on each of 16,000 networks.
Class C	192.0.1.1 to 223.255.254.254	Supports 254 hosts on each of 2 million networks.
Class D	224.0.0.0 to 239.255.255.255	Reserved for multicast groups.
Class E	240.0.0.0 to 254.255.255.254	Reserved.

Internet Protocol hierarchy contains several classes of IP Addresses to be used efficiently in various situations as per the requirement of hosts per network. Broadly, the IPv4 Addressing system is divided into five classes of IP Addresses. All the five classes are identified by the first octet of IP Address.

The first octet referred here is the left most of all. The octets numbered as follows depicting dotted decimal notation of IP Address:



The number of networks and the number of hosts per class can be derived by this formula:

Number of networks	= 2^network_bits
Number of Hosts/Network	= 2^host_bits – 2

When calculating hosts' IP addresses, 2 IP addresses are decreased because they cannot be assigned to hosts, i.e. the first IP of a network is network number and the last IP is reserved for Broadcast IP.
The default subnet mask for Class A IP address is 255.0.0.0 which implies that Class A addressing can have 126 networks (2 7-2) and 16777214 hosts (2 24-2).

Class A IP address format is thus: **0NNNNNNN.HHHHHHHH.HHHHHHHH.HHHHHHHH**

Class B Address:

An IP address which belongs to class B has the first two bits in the first octet set to 10, i.e.

10000000 – 10111111
128 – 191

Class B IP Addresses range from 128.0.x.x to 191.255.x.x. The default subnet mask for Class B is 255.255.x.x. Class B has 16384 (2 14) Network addresses and 65534 (2 16-2) Host addresses.

Class B IP address format is: **10NNNNNN.NNNNNNNN.HHHHHHHH.HHHHHHHH**

Class C Address:

The first octet of Class C IP address has its first 3 bits set to 110, that is:

11000000 – 11011111
192 – 223

Class C IP addresses range from 192.0.0.x to 223.255.255.x. The default subnet mask for Class C is 255.255.255.x. Class C gives 2097152 (2 21) Network addresses and 254 (2 8-2) Host addresses.

Class C IP address format is: **110NNNNN.NNNNNNNN.NNNNNNNN.HHHHHHHH**

Class D Address:

Very first four bits of the first octet in Class D IP addresses are set to 1110, giving a range of:

11100000 – 11101111
224 – 239

Class D has IP address rage from 224.0.0.0 to 239.255.255.255. Class D is reserved for Multicasting. In multicasting data is not destined for a particular host, that is why there is no need to extract host address from the IP address, and Class D does not have any subnet mask.

Class E Address

This IP Class is reserved for experimental purposes only for R&D or Study. IP addresses in this class ranges from 240.0.0.0 to 255.255.255.254. Like Class D, this class too is not equipped with any subnet mask.

Private IP address:

A private IP address is an IP address that's reserved for internal use behind a router or other Network Address Translation (NAT) device, apart from the public.

Private IP addresses are in contrast to public IP addresses, which are public and can not be used within a home or business network.

Sometimes a private IP address is also referred to as a local IP address.

Public IP address:

An Internet Protocol (IP) address that is designated for use in a public domain, such as the Internet. A public IP address is in contrast to a private IP address, which is in an address range designated for use only in a private domain, such as a local area network (LAN). See also domain, Internet, IP, IP address, LAN, and private IP address.

Summary of Ipv4 Classes range and Public & Private IP address:

IP address classes

Class	1st Octet Decimal Range	1st Octet High Order Bits	Network/Host ID (N=Network, H=Host)	Default Subnet Mask	Number of Networks	Hosts per Network (Usable Addresses)
A	1 – 126*	0	N.H.H.H	255.0.0.0	126 ($2^7 - 2$)	16,777,214 ($2^{24} - 2$)
B	128 – 191	10	N.N.H.H	255.255.0.0	16,382 ($2^{14} - 2$)	65,534 ($2^{16} - 2$)
C	192 – 223	110	N.N.N.H	255.255.255.0	2,097,150 ($2^{21} - 2$)	254 ($2^8 - 2$)
D	224 – 239	1110	Reserved for Multicasting			
E	240 – 254	1111	Experimental; used for research			

Note: Class A addresses 127.0.0.0 to 127.255.255.255 cannot be used and is reserved for loopback and diagnostic functions.

Private IP Addresses

Class	Private Networks	Subnet Mask	Address Range
A	10.0.0.0	255.0.0.0	10.0.0.0 - 10.255.255.255
B	172.16.0.0 - 172.31.0.0	255.240.0.0	172.16.0.0 - 172.31.255.255
C	192.168.0.0	255.255.0.0	192.168.0.0 - 192.168.255.255

To configure the IP address in Windows 10, complete the following steps.

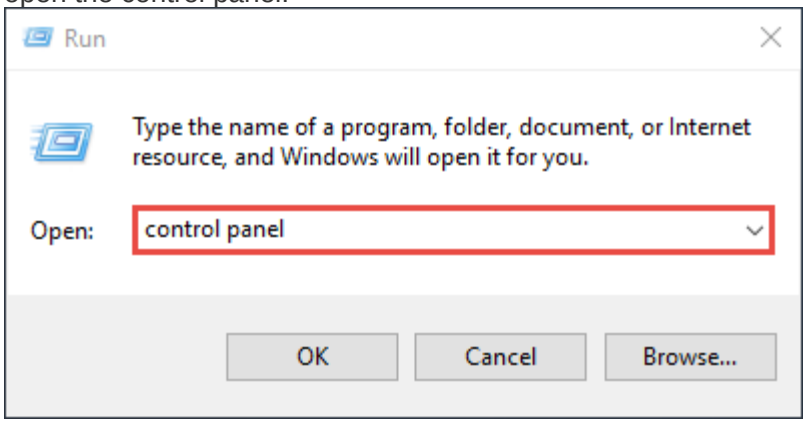
Procedure

1. Click **Start** and type **Control Panel** in the Windows search bar.
2. In the Control Panel window, select **Network and Internet** > **Network and Sharing Center** to view the network information and to set up a connection.
3. Select **Change adapter settings** from the left navigation pane to open the Network Connections window.
4. Right-click the **Ethernet** icon and select **Status** from the drop-down list.
5. Click the **Details** button in the Ethernet Status window.
6. Right-click the **Local Area Connection** icon and click **Properties**.
7. In the Ethernet Properties window, enable the **Internet Protocol Version 4 (TCP/IPv4)** option. The Internet Protocol Version 4 (TCP/IPv4) window opens.
8. Select the **Use the Following IP address** option and enter the values in the **IP address**, **Subnet mask** and **Default gateway**

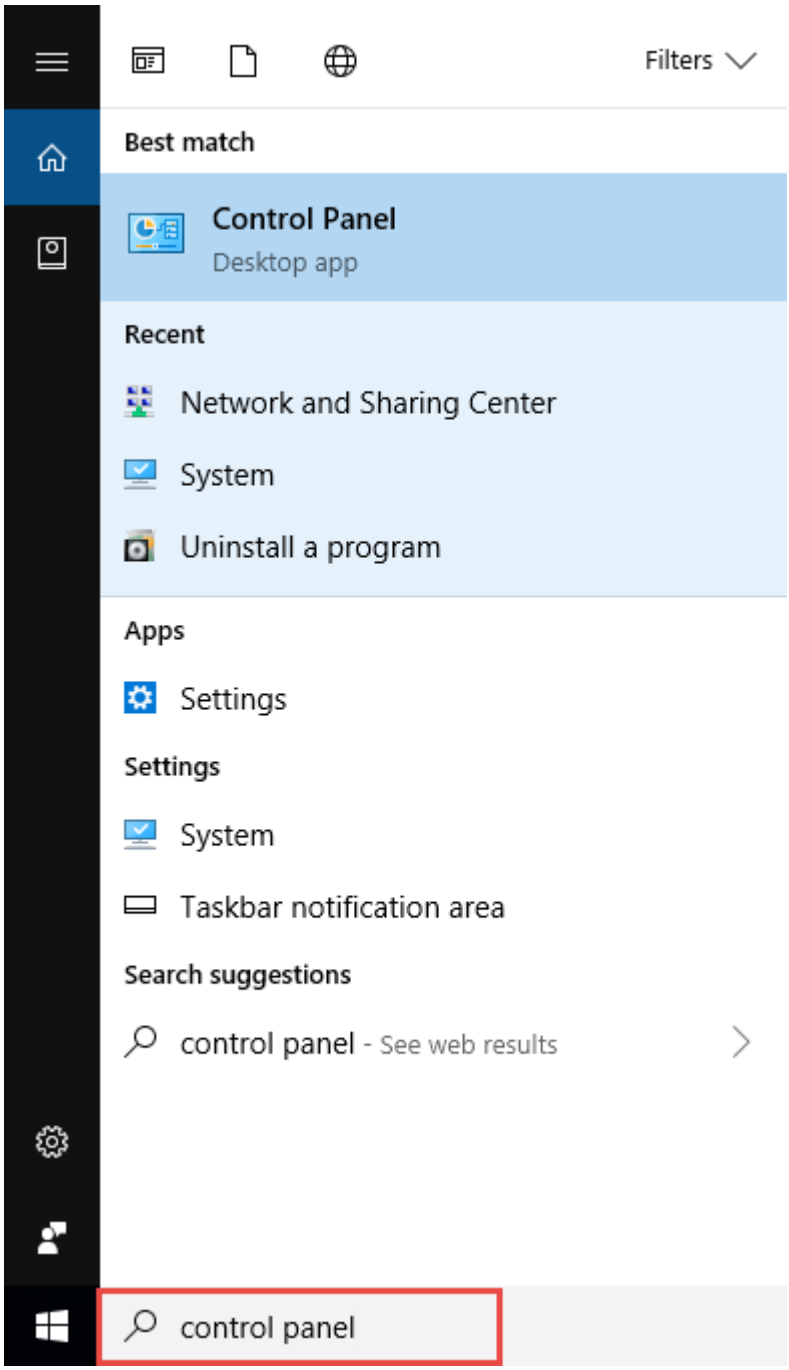
9. Click **OK** to apply the settings.

Step 1: Open the Control Panel

Press "**Windows**  + **R**", then a **Run** box comes out. Input **control panel** and press **Enter** to open the control panel.

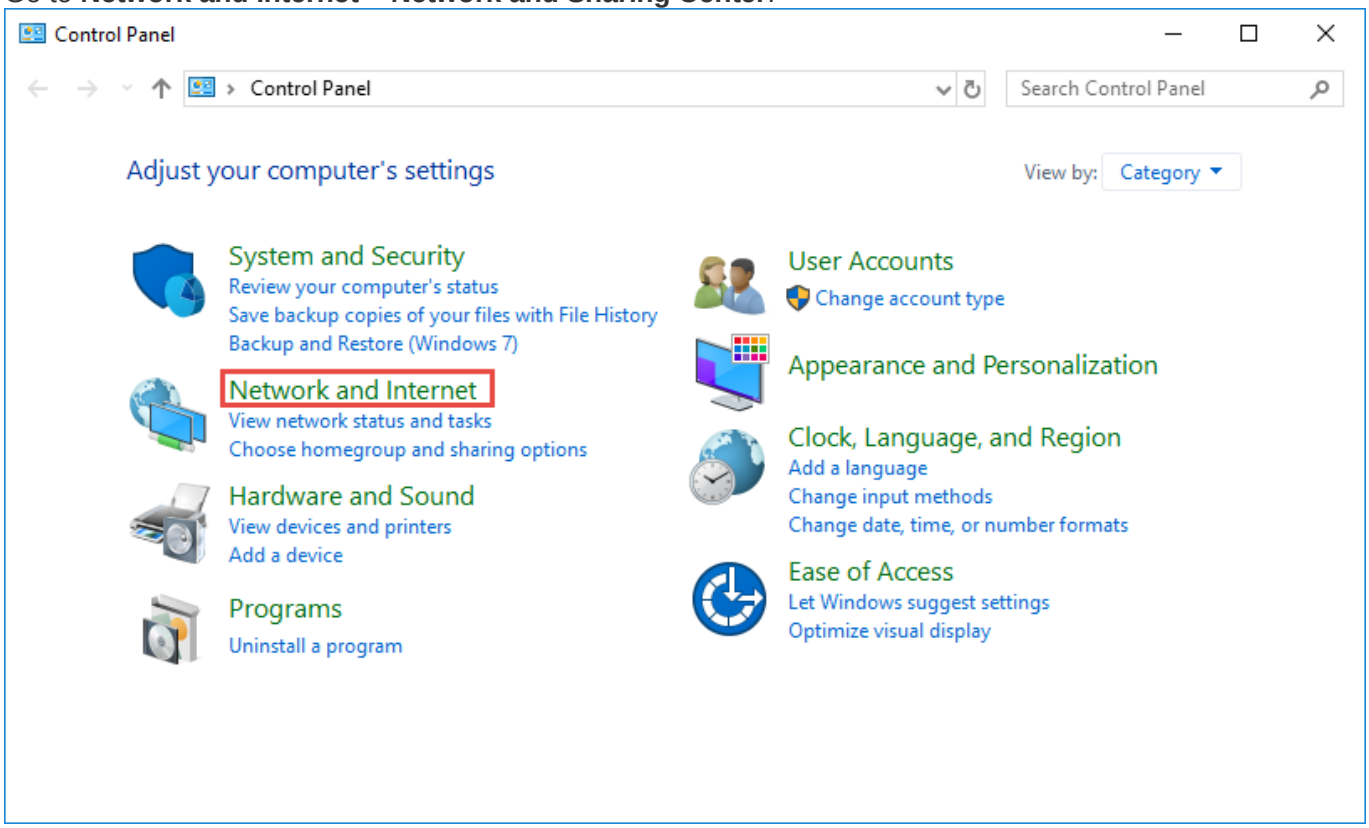


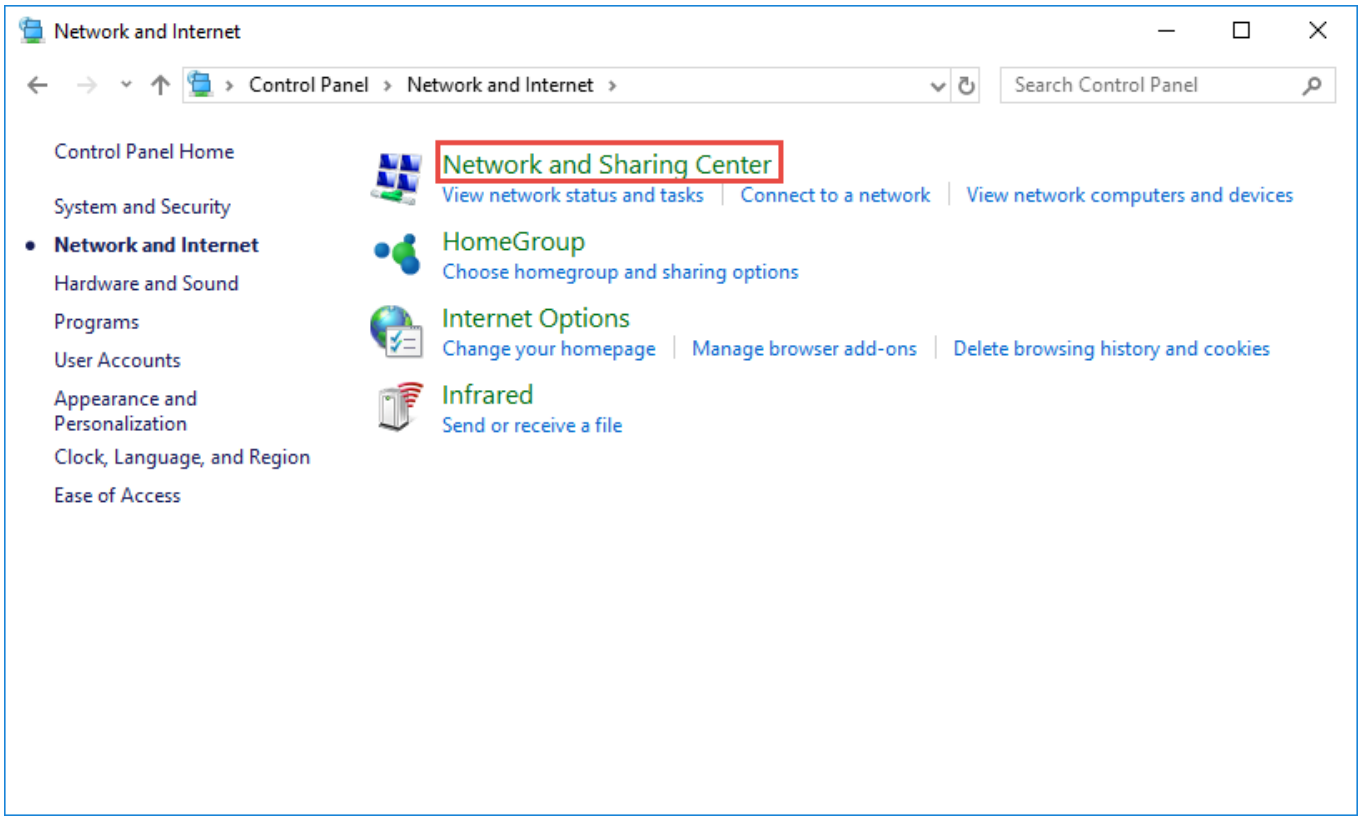
You can also type **control panel** in the search bar at the lower left of the screen and press **Enter** to open the control panel.


Filters
Best match
Control Panel
Desktop app
Recent
Network and Sharing Center
System
Uninstall a program
Apps
Settings
Settings
System
Taskbar notification area
Search suggestions
control panel - See web results
control panel

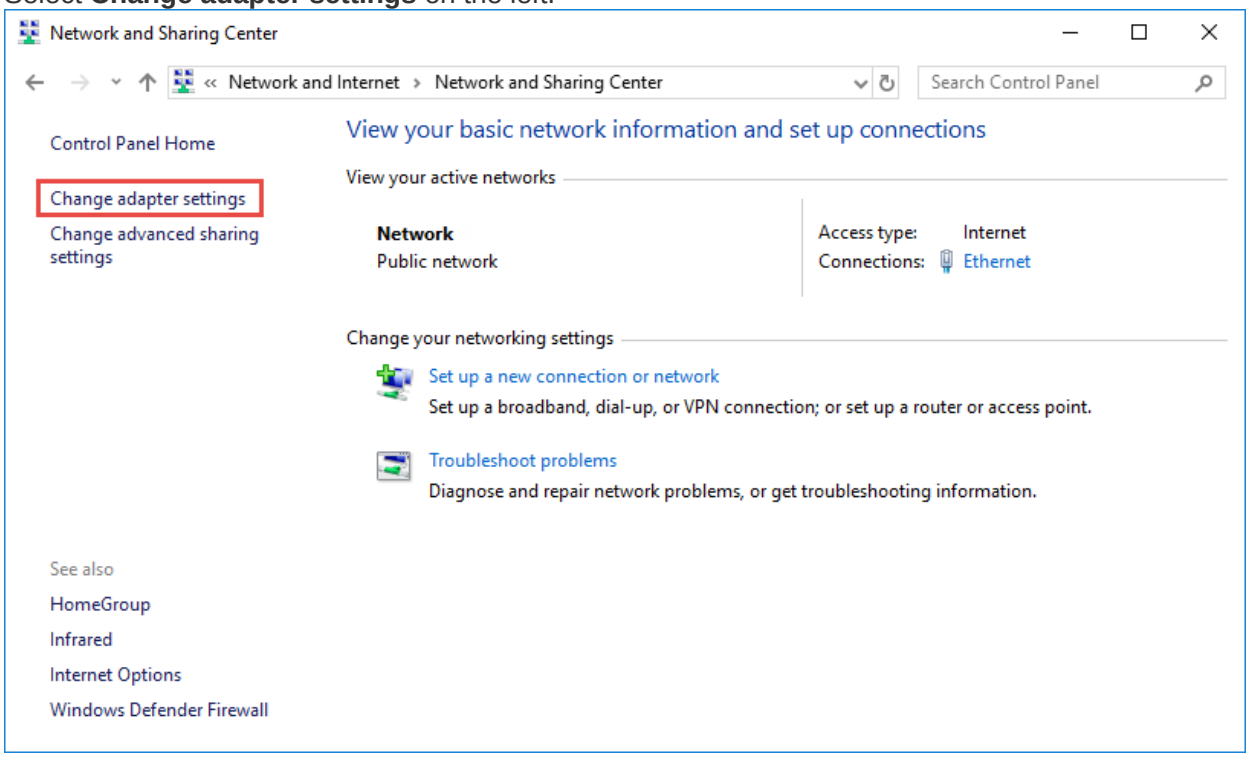
Step 2: Go to Network Connections

Go to **Network and Internet** > **Network and Sharing Center**.



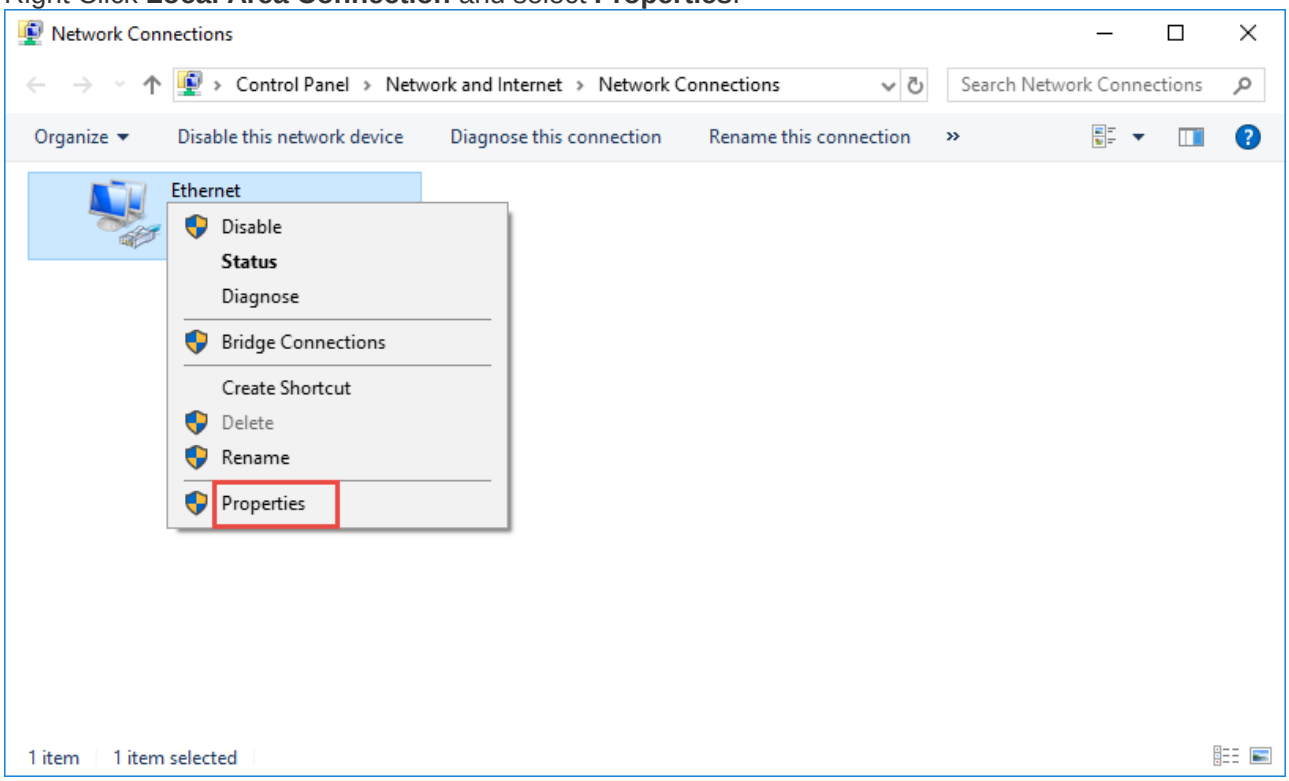


Select **Change adapter settings** on the left.

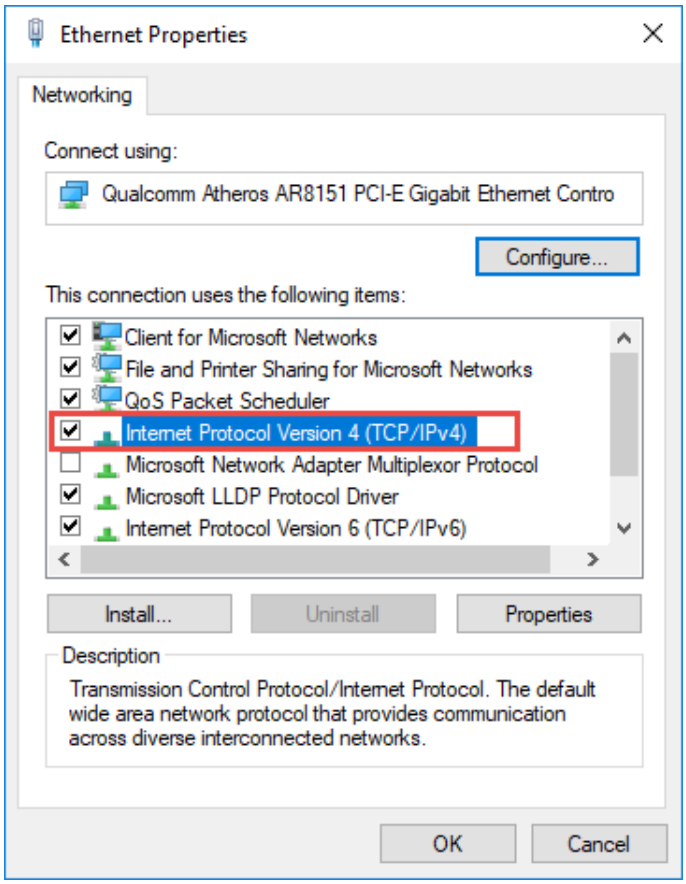


Step 2: Set the IP address

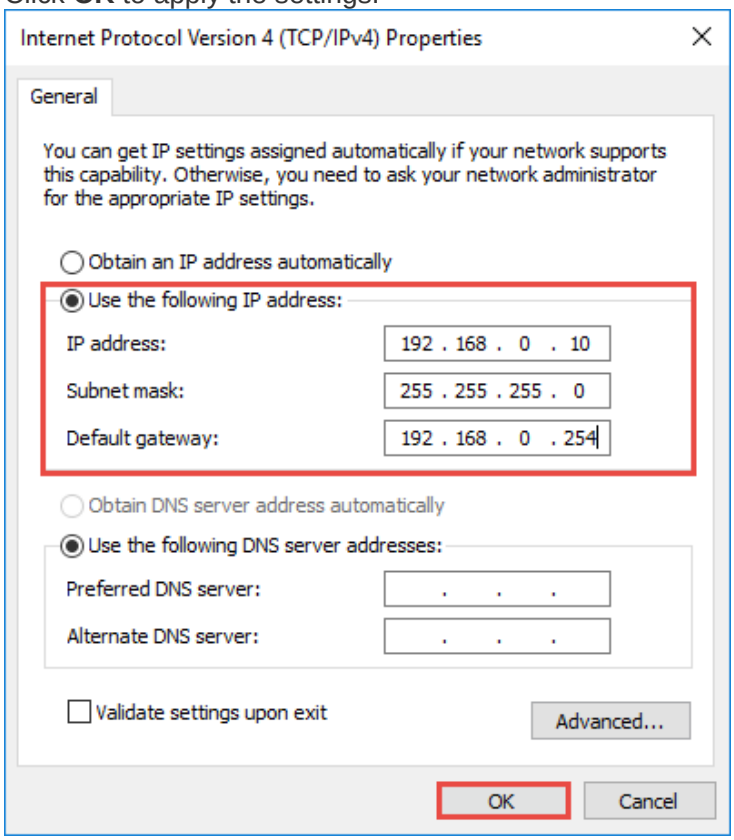
Right Click **Local Area Connection** and select **Properties**.



Then double click **Internet Protocol Version 4 (TCP/IPv4)**.



Select **Use the Following IP address:** and type in the **IP address**, **Subnet mask** and **Default gateway**. Click **OK** to apply the settings.



Review Questions

Question 1: What kind of networks use a Class A or Class B or Class C IP address typically?
Answer:

Question 2: What is the LOOPBACK address?
Answer:

Question 3: How many number of networks and hosts can be represented using the IP address classes??
Answer:

Note: Submit the handout before next lab